

AUN ALI

HEALTHCARE SOLUTION ARCHITECT | ENTERPRISE INTEGRATION | PUBLIC HEALTH TECHNOLOGY

Katy, Texas | (832) 235-5025 | aunaliadenwala@hotmail.com | LinkedIn: linkedin.com/in/aun-a-b9127433 | aunali.github.io/Profile

PROFESSIONAL SUMMARY

Healthcare Solution Architect and IT leader with 25+ years of experience designing, modernizing, integrating, and supporting enterprise healthcare and public health platforms. Deep hands-on expertise across HL7, FHIR, CDA/eICR, ELR, EDI X12, Mirth Connect, Rhapsody, SQL Server, .NET, Azure, Snowflake, Power BI, and API-based integration. Proven record leading mission-critical surveillance systems, international HIS implementations, data-center/cloud migrations, large-scale application modernization, vendor programs, and cross-functional technical teams.

ARCHITECTURE & LEADERSHIP HIGHLIGHTS

Enterprise Architecture Healthcare application architecture, integration patterns, modernization roadmaps, cloud/data architecture, scalability, reliability, and technical governance.	Healthcare Interoperability HL7 v2, FHIR, CDA, eICR, ELR, ADT, ASTM, EDI X12, REST/SOAP APIs, Mirth Connect, Rhapsody, BizTalk, nHapi.
Data & Analytics SQL Server, Snowflake, Azure Data Factory, dimensional modeling, data marts, ETL, Power BI, Crystal Reports, data quality and validation.	Technical Leadership Product ownership, stakeholder discovery, RFP support, vendor management, architecture reviews, implementation planning, mentoring, production support.
Application Engineering C#, .NET, ASP.NET, WPF, Windows Services, Web APIs, XML/JSON/XSLT, VB.NET/VB6, SQL/T-SQL.	AI & Automation RAG/knowledge assistants, AI-enabled document processing, workflow automation, configurable alerting, and healthcare/public health AI solution evaluation.

SELECTED IMPACT

- Lead a highly configurable public health disease surveillance platform serving 1,000+ users across multiple programs and disease workflows.
- Oversee an ESSENCE syndromic surveillance environment receiving approximately 300,000 HL7 messages daily from regional healthcare facilities.
- Rebuilt approximately 22 hospital information-system modules using Microsoft Windows/.NET technologies, including ADT, RIS, and CSSD/CICS workflows.
- Led or supported healthcare implementations across the United States, Pakistan, Afghanistan, and Kenya, including user training, infrastructure readiness, application configuration, and go-live support.
- Implemented EDI X12 warehouse-management integrations for Willis-Knighton Health System, including 850 Purchase Orders, 810 Invoices, 856 Ship Notices, and UB-04 to X12 837 claim conversion.
- Supported Point-of-Use and healthcare supply-chain solutions across 20 U.S. Department of Veterans Affairs hospital sites.

PROFESSIONAL EXPERIENCE

City of Houston - Houston Health Department | IT Lead - Applications

Houston, Texas | Nov 2020 - Present

- Lead architecture, modernization, implementation, and production support for mission-critical public health applications, integration platforms, reporting solutions, and data-exchange initiatives.
- Provide technical leadership for the Health Department Surveillance System, a configurable platform supporting 1,000+ users, disease-specific workflows, questionnaires, investigations, and reporting.
- Lead healthcare interoperability solutions for HL7, ELR, eICR/CDA, ADT, immunization, laboratory, and public health data exchange using Mirth Connect, Rhapsody, SQL Server, .NET, APIs, XML, and JSON.
- Serve as Product Owner for Laboratory Information System modernization, partnering with laboratory stakeholders on requirements, RFP development, solution evaluation, and replacement planning.
- Implemented TB PACS modernization and migration from a legacy OPAL environment to a cloud-based radiology platform; provide technical leadership for TB case-management modernization and data migration.
- Lead ESSENCE infrastructure modernization and production operations for a syndromic surveillance platform receiving approximately 300,000 HL7 messages daily; coordinate platform upgrades with the vendor.
- Architect and develop eICR processing capabilities for retrieval, CDA/XML transformation, validation, search, presentation, configurable mapping, and public health workflows.
- Led data-center modernization efforts, including application rebuilds, physical-to-virtual migrations, database migrations, interface-engine migrations, configuration validation, and production cutover.

- Design and support enterprise analytics and data modernization using SQL Server, Azure Data Factory, Snowflake, dimensional modeling, and Power BI, including immunization reporting and data-quality validation.
- Evaluate and implement AI-enabled solutions such as knowledge assistants/RAG, document processing, workflow automation, and configurable public health alerting while coordinating vendors, security, infrastructure, and program leadership.

The Shams Group, LLC | Software Engineer

Dallas, Texas | May 2018 - Nov 2020

- Designed and implemented healthcare supply-chain and interoperability solutions using EDI X12, HL7, Mirth Connect, SQL Server, C#, .NET, WPF, and REST/SOAP services.
- Implemented EDI X12 warehouse-management integration for Willis-Knighton Health System, including 850 Purchase Orders, 810 Invoices, 856 Ship Notices/Manifests, and an EDI data mart for operational reporting.
- Designed and implemented UB-04 institutional claim transformation to ANSI X12 837 electronic healthcare claims.
- Automated receiving and point-of-care workflows and enhanced handheld/UWP applications supporting warehouse and inventory operations.
- Developed .NET/WPF applications including ZPL barcode/label printing and HL7-based integration between reporting systems and healthcare vendors; participated in FHIR-based EHR integration design.

Shipcom Wireless Inc. | Application Developer

Houston, Texas | Mar 2015 - Apr 2018

- Developed and supported U.S. Department of Veterans Affairs Point-of-Use healthcare supply-chain solutions and implementations across 20 VA hospital sites.
- Designed HL7 integrations for VA VISN 10 RTLS and healthcare asset-management workflows using Mirth Connect, C#, nHapi, SQL Server, and WPF.
- Created a .NET RPC Broker and Windows/Web Service components integrating VA Vista/IFCAP/GIP legacy systems with modern supply-chain applications.
- Integrated supply-chain systems with Omnicell, Pyxis, Par Excellence, and other smart-cabinet platforms; automated RFID/LogiTag inventory workflows using SQL stored procedures and the Catamaran framework.

Dynamed Clinical Research L.P. | Systems Analyst

Tomball, Texas | Feb 2015 - Mar 2015

- Designed and implemented machine-to-system integrations using HL7 and ASTM protocols for clinical research workflows and laboratory devices.

The Aga Khan University Hospital | Software Designer / Senior Programmer Analyst / Programmer Analyst

Karachi, Pakistan | Jun 2005 - Dec 2014

- Designed, developed, implemented, and supported a broad portfolio of hospital information-system solutions spanning ADT, RIS, oncology, medical records, outpatient management, CSSD/CICS, ER triage, immunization, labor room management, physician portals, and scheduling.
- Rebuilt approximately 22 MEDITECH-related hospital modules using Microsoft Windows/.NET technologies, including core patient, ADT, RIS, and CSSD/CICS workflows.
- Led implementation of Hospital Information Systems at FMIC Kabul, Afghanistan, including hardware planning, application setup, HIS/ADT/OPMS configuration, user training, go-live preparation, and production support.
- Implemented and supported hospital information systems at Aga Khan University Hospital Nairobi, Kenya, and multiple hospitals in Pakistan, adapting workflows, applications, and training to local operational needs.
- Implemented Varian/ARIA Oncology and managed integrations with enterprise vendor solutions including Nuance/PowerScribe, Roche, and other clinical systems using HL7, ASTM, BizTalk, and Mirth Connect.
- Designed the Expected Transfer Admission (ETA) solution for the E-Health initiative and received an Aga Khan University internal award for successful development and implementation.
- Led small development teams and participated in full SDLC/RUP activities including requirements, architecture/design, development, testing, documentation, implementation, and production support.

2B Technologies | System & Software Support Engineer

Karachi, Pakistan | Sep 2004 - Jun 2005

- Developed IVR functionality for Saudi Pak Bank and provided software/system support for enterprise call-center solutions.
- Administered SQL Server 2000 call-center databases for multiple clients, including database creation, maintenance, reporting, troubleshooting, and production support.

SELECTED ARCHITECTURE PROJECTS

- **Public Health Disease Surveillance:** enterprise application leadership for a configurable surveillance platform supporting 1,000+ users, multiple programs, disease workflows, questionnaires, investigations, integrations, and reporting.
- **ESSENCE Syndromic Surveillance:** infrastructure migration, production architecture, vendor coordination, and upgrade planning for a high-volume platform receiving approximately 300,000 HL7 messages daily.
- **eICR / Electronic Case Reporting:** configurable architecture for CDA/eICR ingestion, XML transformation, validation, search, presentation, alerting, and public health workflows.

- **Immunization Data Modernization:** SQL Server, Azure Data Factory, Snowflake dimensional modeling, automated data pipelines, validation, and Power BI reporting.
- **LIS Modernization:** product ownership and solution planning for replacement of a legacy Laboratory Information System, including requirements, RFP support, integration planning, and vendor evaluation.
- **TB Technology Modernization:** TB PACS/radiology migration, TB case-management modernization, large-scale data migration, and integration of ELR/eICR workflows.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master of Computer Science - SZABIST, Karachi, Pakistan | 2012 | CGPA 3.36

Bachelor of Engineering in Information Technology - Hamdard University, Karachi, Pakistan | 2005 | CGPA 3.46

Training: Health Level 7 (HL7), Nuance PowerScribe Voice Recognition, Agile Project Management, CBAP

TECHNICAL ARCHITECTURE STACK

Interoperability & Standards: HL7 v2, FHIR, CDA, eICR, ELR, ADT, ASTM, EDI X12, SOAP, REST, HTTP, RPC, XML, JSON, XSLT

Integration Platforms: Mirth Connect, Rhapsody, BizTalk Server, Interfaceware, nHapi, Postman, Fiddler, HL7Edit

Cloud, Data & Analytics: Azure, Azure Data Factory, Snowflake, SQL Server, Oracle, MySQL, T-SQL, PL/SQL, ETL, dimensional modeling, Power BI, Crystal Reports

Development: C#, .NET, ASP.NET, WPF, Windows Services, Web API, VB.NET, VB6, C++, JavaScript, UML, Visual Studio

Healthcare Domains: Public Health Surveillance, EHR/HIS, LIS, PACS/RIS, Oncology, Immunization, Syndromic Surveillance, Laboratory Reporting, Healthcare Supply Chain, Claims/EDI